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The Editor-in-Chief
Simulation Modelling Practice and Theory

Dear Editor,

I submit the manuscript "Verified coupled simulation and uncertainty analysis of floating photovoltaic lifetime performance in humid and saline environments" for consideration as a research article in *Simulation Modelling Practice and Theory*.

Floating photovoltaic plants are usually credited with higher yield because water cools the modules. In hot, humid and saline environments, the same water body also accelerates soiling and module degradation. Deciding between floating and land-based PV is therefore a simulation problem with three characteristic difficulties. The mechanisms are coupled through a shared microclimate, they act on time scales from hours to decades, and several governing parameters are unmeasured at a candidate site.

The manuscript contributes to simulation practice in the following ways:

- **Coupled multi-time-scale model.** A single microclimate drives thermal, optical, soiling and four degradation mechanisms in one model, which resolves trade-offs that sub-models fed with independent inputs would miss.
- **Verification and validation.** The V&V assessment follows Sargent's framework. All established sub-models are verified code-to-code against the pvlib reference library and agree to within 3×10^{-12}. Every model component is classified by its evidential basis, the weather generator is checked for consistency with its input climatology, and a field-validation protocol is derived from the sensitivity results.
- **Designed uncertainty experiments.** Weather variability and parameter uncertainty (28 parameters) are separated. A Monte Carlo analysis is checked for convergence up to 5,000 samples (shifts $\leq$0.28 percentage point), and a Sobol variance decomposition shows near-additive behavior with few influential parameters.
- **Methodological finding.** Weather variability alone suggests a precise freshwater advantage of $\pm$0.1 percentage point. Parameter uncertainty shows that this advantage is conditional: it is positive in 75% of samples. The saline penalty, in contrast, is robust (median -5.1%).
- **Reproducibility.** The open-source implementation, unit tests, parameters, seeds and a single reproduction script regenerate every number, table and figure.

The results are presented explicitly as model-based scenario predictions; operational validation against field data remains future work, for which a measurement protocol is proposed.

This manuscript has not been published and is not under consideration for publication elsewhere. As stated in the manuscript, generative AI (Claude, Anthropic) was used to develop the simulation code, run the experiments, and draft and revise the text. The author reviewed and edited all content and takes full responsibility for it. There are no competing interests to declare.

Thank you for considering this submission.

Sincerely,

Subrahmanyam Tanala